

MASTER OF BUSINESS ADMINISTRATION

Report on Industry Visit

Firm	Triveni Engineering & Industries Ltd.
Address	Belagola Industrial Area, Metagalli, Mysuru, Karnataka
Nature of the Firm:	Manufacturing
Date of Visit	24-02-2026
Scheduled time of visit	10:00 am to 12:30 pm

I. Objectives of the Program:

The visit to Triveni Engineering & Industries Limited (PTB), Mysuru, aimed to achieve these primary goals:

- **Theory-to-Practice Bridge:** Connect academic concepts in Operations Management, Supply Chain, and Material Handling with live industrial applications.
- **Full Process Observation:** Examine the complete lifecycle of high-speed gear production, from initial design through heat treatment to final precision assembly.
- **Global Quality Insights:** Explore how international benchmarks like AGMA and API are applied in a major engineering operation.
- **Efficiency Practices:** Observe real-time implementation of 5S, Lean principles, and integrated supply chain coordination on the shop floor.
- **Deliver hands-on exposure** to large-scale manufacturing and cutting-edge technologies.
- **Illustrate** the synergy between engineering precision and management strategies in production planning and quality assurance.
- Link theoretical classroom knowledge to practical industry scenarios.

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II. Introduction to Triveni Engineering & Industries Ltd.

Triveni Engineering & Industries Limited (TEIL), founded in 1932, stands as a dynamic Indian multinational with deep expertise across sectors. Renowned as India's top integrated sugar and ethanol producer, it also excels in engineering, particularly through its Power Transmission Business (PTB).

Headquartered in Noida, the company's advanced engineering hub thrives in Mysuru, Karnataka.

The Mysuru Facility: A Global Gear Manufacturing Powerhouse
Launched in 1976, this 10.5-acre cutting-edge plant has grown into one of the world's leading centers for high-speed gear production.

- **Regional Dominance:** Holds over 80% market share for high-speed gears in South and Southeast Asia.
- **Worldwide Impact:** Supports more than 50,000 MW of global gear installations, serving 600+ clients across 70+ countries.
- **National Significance:** Key supplier to the Indian Navy for propulsion gears and defense systems, advancing self-reliance goals.

II a. Core Business Segments at Mysuru

The plant focuses on three vital areas, offering MBA students clear views of diversified operations:

1. **Gears & Gearboxes:** Custom design and production of ultra-high-speed gears (up to 70,000 RPM) for turbines, compressors, and industrial pumps.
2. **Aftermarket Solutions:** Overhauls, retrofits, and spares for 90+ international brands, fostering lifetime customer loyalty and revenue.
3. **Defence & Marine Division:** Critical components for naval vessels and coast guard units, aligning with "Atmanirbhar Bharat."

III. Brief Overview of the Visit:

Thirty-five MBA students, led by a faculty member, toured Triveni Engineering & Industries Ltd., Mysuru. The group was greeted at the factory gate by the Training & Development Officer, who delivered a warm welcome and overview of the company's legacy, operations, and industry contributions. The program kicked off with a management presentation, then shifted to a guided walkthrough of the

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expansive 10.5-acre site.

- **Shop Floor Highlights:** Students examined advanced zones like the 7-axis CNC grinding cells and dedicated heat treatment facilities.
- **Testing Highlights:** A standout was the assembly bay, where enormous gearboxes (for naval and power sector use) endure high-speed simulations up to 70,000 RPM.

The Gearbox Manufacturing Process

Triveni's Mysuru plant uses a high-precision pipeline to produce durable gearboxes for extreme torque and conditions. Premium steels arrive via vendor-managed inventory, then undergo forging, automated flaw inspections, blanking, CNC turning, and hobbing for precise teeth. Heat treatment achieves 62 HRC surface hardness with a ductile core, followed by CBN grinding to AGMA 12-14 tolerances. Robotic machining creates aligned casings; clean-room assembly and no-load dyno tests ensure quality. Finished units receive coatings, crating, and GPS-tracked global shipping.

Learning Outcomes

For MBA students, the visit yielded these vital insights:

1. **Quality as a Competitive Advantage:** API/AGMA compliance isn't mere regulation—it's a powerhouse for premium positioning and export success.
2. **Product Lifecycle Management:** With 20–30-year lifespans, gearboxes demand proactive aftermarket ecosystems for sustained profitability.
3. **Work Culture:** "Safety First" protocols and 5S rigor on the floor underscore culture's role in zero-accident, high-output environments.
4. **Operations Management:** Tools like value stream mapping and SMED (Single-Minute Exchange of Die) maximize throughput and worker efficiency.
5. **Total Quality Management:** Multi-tiered systems, including poka-yoke error-proofing and ISO 9001 audits, embed quality throughout.
6. **Supply Chain Management Concepts:** Seamless flows in procurement (e-auctions), inventory (JIT/Kanban), handling (forklifts/AGVs), and logistics revealed end-to-end optimization.

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Conclusion

The industrial visit to Triveni Engineering & Industries Limited (PTB), Mysuru provided MBA students with valuable real-world exposure to large-scale, precision-driven manufacturing. The visit successfully bridged classroom concepts in Operations Management, Supply Chain, Quality Management, and Lean Practices with live industrial applications.

Students gained practical insights into high-speed gear production, global quality standards (AGMA/API compliance), advanced manufacturing technologies, and integrated supply chain systems. The experience highlighted how engineering excellence, strong work culture, safety practices, and strategic management together create sustainable competitive advantage. Overall, the visit enhanced students' understanding of how theoretical knowledge is translated into efficient production planning, quality assurance, and long-term business success in a global engineering enterprise.

PHOTO OF THE VISIT

